

Use peak-detect mode to calculate small pulses

When the sampling rate of a digital oscilloscope is not high enough to fully capture details of a signal, you can use peak-detect mode to help capture the envelope or narrow-pulse information that may be lost when using a normal-acquisition mode. To know the difference between normal mode and peak-detect mode, use an arbitrary waveform generator to output a signal. For example, the signal could be 1kHz in frequency, with 5V peak-to-peak amplitude and 10ns pulse width. Connect this signal to channel one of your oscilloscope and adjust the time base of the oscilloscope to 1ms/div, vertical scale to 1V/div and memory depth to 140kpt.

The acquisition, or acquire mode, on your oscilloscope is usually set to normal mode and in that mode, there are some pulses with varying amplitudes that are seen, but the display is not stable. By changing the mode to peak-detect mode, the amplitude of pulses is fixed and a stable view is available. Therefore to accurately measure narrow pulses using peak-detect mode on your scope, first configure the vertical and time-base scale to capture the signal. Then, choose peak-detect mode to get the exact information about narrow-pulse or peak features within the signal.

Use ground spring probes to achieve signal and noise quality

Most common factors that need to be tested for verifying the quality of every design, especially embedded systems, are data-transmission rate, signal quality and timing specifications. It is important to analyse the overall accuracy and system-level timing delays of an embedded design. Another common issue with testing and verification is noise, which can come from a variety of sources, including electromagnetic

Some Mid-range Digital Oscilloscopes in the Market Today		
Brand	Model	Top features
GW Instek	GBS-1000 series 	• 200MHz/100MHz/70MHz bandwidth range • 1GSa/s real-time sampling rate and 25k point memory depth • 25GSa/s equivalent time-sampling rate • 14.5cm (5.7-inch) TFT LCD display
Keysight (formerly Agilent)	DSO9064A 	• 600MHz bandwidth across all four analogue channels • Three instruments in one: scope, logic analyser and protocol analyser • 38.1cm (15-inch) XGA display with debug and compliance software applications
Pico Technology	PicoScope 3400 	• Bandwidths 60MHz to 200MHz, four channels and external trigger • Up to 128MS buffer size with advanced digital triggers • 1GS/s real-time sampling, up to 10GS/s repetitive sampling • Built-in spectrum analyser and function generator
Rigol	MSO/DS2000A series 	• Bandwidths 70MHz to 300MHz, maximum sample rate 2G Sa/s • Memory depth up to 14Mpt (standard), 56Mpt (optional), waveform capture rate up to 50,000wfs/s • Up to 65,000 frames hardware based real-time waveform record, replay and analysis function (standard) • Built-in two channel waveform generator (DS2000A-S) 20.3cm (8-inch) TFT (800x480) WVGA
Rohde & Schwarz	R&S RTE1054 	• Four channels, 500MHz bandwidth, acquisition memory of 10/40 Msample, upgradable up to 50/200 Msample • 77 measurement functions and two cursor sets, four maths waveforms • 26.42cm (10.4-inch) LC TFT colour touch-screen, 1024x728 pixel (XGA)
Tektronix	DPO5054B 	• 500MHz analogue bandwidth, five GS/s sample rate, 25M point - 125M point record length • FastAcq with digital phosphor display • Built-in analysis tools
Yokogawa	DLM2000 series 	• Bandwidths 200MHz to 500MHz and memory up to 250Mpt • Real-time filter with optimum noise reduction supports a wide range of frequencies (from 8kHz to 200MHz) • Zooms into two different points, waveform zoom and search functions, with dedicated zoom keys

interference (EMI), bandwidth problems and bad grounding.

For noise-related issues, alligator clip ground strap, which connects on the probe, is normally used to establish a ground connection. Although connections are made correctly, noise could still be an issue, in which case it would be advisable to use the ground spring instead. Comparatively, the ground spring connects closer to the tip of the probe and significantly reduces

the loop area of the connection, thereby improving the noise and signal quality.

If ground noise still persists, detaching your oscilloscope from ground could help. By isolating the rest of the device or the device under test, ground loops are eliminated.

Use precision probe for measurements at high bandwidth

Oscilloscopes have reached very high bandwidth levels today. For instance,